Vascular Cut Tool — User Guide

Turn a closed vascular surface mesh into an **open** mesh — with clean planar inlets and outlets — by placing one cut per opening along the vessel centerline. Works for branched and unbranched aortas.

Getting started

You have been given a single file, **vascular-cut.zip**. Here is how to get from that to a running tool.

1. **Unzip it.** Copy **vascular-cut.zip** onto your computer and unzip it (on Windows: right-click → *Extract All*). You get a folder called **vascular-cut**.
2. **Launch it.**

- **Windows:** open the folder and double-click **run.bat**. If a blue "Windows protected your PC" box appears, click *More info* → *Run anyway* (it is your own lab's script, not from the internet).
- **macOS / Linux:** open a Terminal in the folder and run **./run.sh** (the first time only, run `chmod +x run.sh` once).

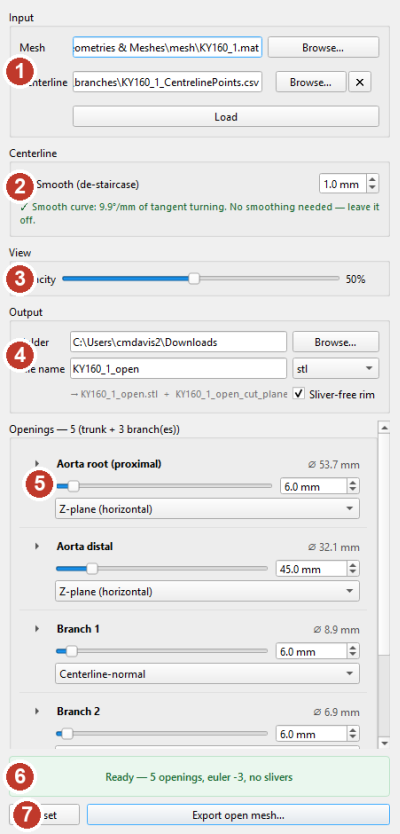
3. **Wait for the first-run setup.** The very first launch builds the tool's environment automatically — a few minutes, with a progress log scrolling by. **Every launch after that opens in seconds.** When the window appears, you are ready — see *The interface* on the next page.

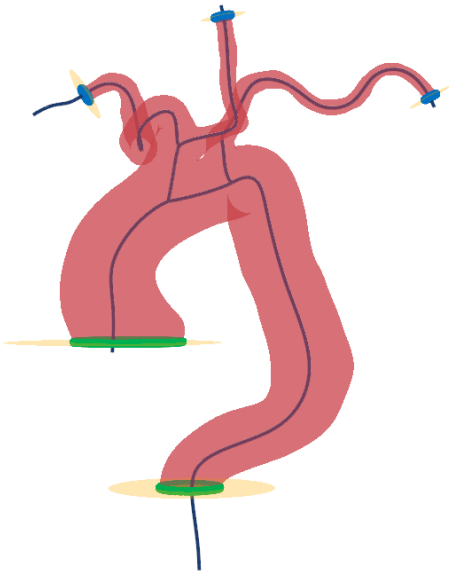
> **Before you start — one thing must already be on the computer:** either **Anaconda / Miniforge** (conda) **or Python 3.10, 3.11, or 3.12**. Most lab machines already have one of these; the launcher finds it automatically. If you have neither — or are not sure — ask your IT support or whoever sent you this tool. You also need an **internet connection the first time** you launch, so it can download its components.

Nothing to install by hand, no admin rights needed for the tool itself — the launcher sets up everything inside the folder. Full technical setup notes (conda / pip alternatives) are in **README.md** in the same folder.

The interface

The control panel is on the left; the 3-D view is on the right.





1	**Input** — browse to a surface **Mesh** (.mat / .stl / .ply / .obj) and, to cut it, a **Centerline** (.csv). Leave the centerline blank to just view a mesh. Click **Load**.
2	**Centerline** — the readout grades the centerline. If it is a voxel “staircase”, tick **Smooth**. Already-smooth centerlines are flagged and left untouched.
3	**Opacity** — make the mesh see-through so you can watch the centerline and cut planes inside it.
4	**Output** — folder, file name, and format for the exported mesh (remembered between sessions).
5	**An opening** — one row per cut: slider + exact-mm box, cut-mode dropdown, and the **expander arrow** (far left of the row) for advanced tilt / plane-size controls. The number on the right is the current opening diameter.
6	**Status bar** — **green** = a valid, clean cut; **red** names the problem and the fix. Export is blocked while it is red.
7	**Reset** and **Export open mesh…**

The workflow

Five steps, start to finish.

1 Launch it

Double-click **run.bat** (Windows) or run **./run.sh** (macOS / Linux). The first launch sets up its environment (a few minutes); every launch afterwards is instant.

2 Load a case

Browse to your mesh and centerline (callout 1) and click **Load**. The sample aorta ships with the tool — `Geometries & Meshes/mesh/KY160_1.mat` with `KY160_1_CentrelinePoints.csv` — so you can try it immediately. The tool works out the vessel tree, labels the openings (aorta root, distal, and each branch), and picks a sensible cut mode for each, all from the geometry.

3 Place each cut

Every opening has its own row (callout 5). Set how far back from the vessel tip the cut sits:

- **Drag the slider** for a feel — the cut plane follows live in the 3-D view; the mesh re-cuts when you let go.
- **Type an exact number** in the mm box when you need a precise distance.

4 Fine-tune (only if needed)

Click the **expander arrow** at the far left of a row to reveal advanced controls for that cut:

- **Tilt** — lean the cut plane off its default orientation.
- **Plane size** — how far the cut reaches around the vessel. *Shrink it* if one cut ever punches two holes on a very tortuous branch.
- **Reset plane** — return that opening to automatic.

5 Check, then export

Watch the status bar (callout 6). When it is green, set an output folder, file name, and format (callout 4), then click **Export open mesh…**. You get two files:

- **<name>.stl** (or .ply / .obj) — the open mesh.
- **<name>_cut_planes.json** — a record of every cut: its plane, offset, tilt, cut mode, and resulting diameter.

The exported mesh is your original surface with the ends removed — the geometry is otherwise untouched (no smoothing or remeshing).

Measuring distances and diameters

Optional tools for inspecting a case before or after cutting. They only measure — they never move a vertex or change the mesh.

Turn it on

Tick **Measure along branch (hover, click to pin)** in the **View** panel. A centerline must be loaded. Untick it to hide the markers again.

Hover — distance to the branch node

Move the cursor over the vessel. A live readout gives the distance, in millimetres, measured **along the centerline** from that spot back to the node the branch springs from — for a trunk, the nearest junction; for an unbranched vessel, the nearer end. A dot marks the point on the centerline and the measured stretch of centerline is highlighted.

Click — pin a mark

Click to pin that spot as a mark with its distance. Marks stay put — and survive re-cuts — until you press **Clear marks**. With two or more marks down, the panel also reports the distance **along the centerline** between the last two: that is how you measure a length of vessel between two chosen points.

Live diameter while you place a cut

While you drag a cut's slider, its readout shows the vessel diameter at the plane's current position, updating live. A real cross-section is rarely a perfect circle, so this is the **equivalent-area diameter** (the diameter of a circle of the same area); when the section is noticeably oval the min–max width is shown alongside — e.g. it may read **54.3 mm**, or **54.3 (51–59)** when oval. Hover the number for the full breakdown (area-equivalent diameter, caliper widths, and circularity).

The dots sit on the **centerline** — where each distance is measured from — not on the surface under the cursor.

Reference

Cut modes

- **Z-plane (horizontal)** — a level cut, matching how aorta ends are usually trimmed in segmentation. Auto-picked for trunk ends whose existing cap is near-horizontal.
- **Centerline-normal** — perpendicular to the vessel. Auto-picked for branch tips, and the safe choice anywhere the vessel runs at an angle.

Change any opening's mode with its dropdown, and rename any opening by editing its name — the auto-labels are only a starting point.

Centerline smoothing

The cut plane's orientation comes from the centerline's direction. A jagged "staircase" centerline (e.g. a raw voxel path) makes the plane swing wildly as you move the slider. The **Smooth (de-staircase)** box fits a spline through the points to fix this — without moving the line off-centre. It is **suggested automatically only when needed**; smooth centerlines (e.g. Simpleware / Scan-IP) are left exactly as supplied.

How fine should the input mesh be?

The cut is limited by your **narrowest** vessel, never the wide trunk. As a target edge length for the surface mesh, use about **half the radius of the smallest vessel you intend to cut**. For aortic-arch branches (~3–4 mm radius) that is roughly a **1.5–2 mm** edge; the trunk itself can be much coarser. Meshes exported from Scan-IP / Simpleware are usually already this fine.

Edge length on the smallest vessel you cut	Cross-sectional (flow) area	Result
≤ ½ × its radius	within ~3%	recommended
≤ ~0.8 × its radius	within ~5%	acceptable minimum
> ~2 × its radius	10–35% too small	openings tear — avoid

Two notes: as a mesh coarsens it is the most **tortuous** branch that tears first, not the thinnest; and the tool **refuses a torn opening** (the status bar turns red), so a too-coarse mesh fails safe rather than exporting a bad inlet.

What the status bar tells you

If it says…	It means…	Do this
cutting *along* the vessel (large angle)	a Z-plane is grazing the vessel lengthwise, so the opening would be torn, not round	switch that opening to **Centerline-normal**
the opening is not flat / rim strays off the plane	the cut tore instead of slicing cleanly	move the cut, or change its mode
cuts have met / cut past each other	two cuts on one vessel have collided (unbranched case)	pull one of them back
trimmed …% of the way down its vessel	the cut is very close to the junction it springs from	pull it back a little

Troubleshooting

- **The 3-D window is black.** The view needs a real graphics (OpenGL) context. Over Remote Desktop or a headless session it may not get one — run it on the machine directly, or enable 3-D acceleration in your remote client.
- **“Could not load…” a centerline.** The CSV isn’t a recognised export. The tool reads Simpleware spline exports and cl.py’s `save_centerline_csv` output.
- **A .mat mesh won’t load.** It must contain `<name>_surface_vertices` and `<name>_surface_triangles` (Simpleware’s surface format).

Full setup and install notes are in **README.md** in the project folder.